RAG Learning

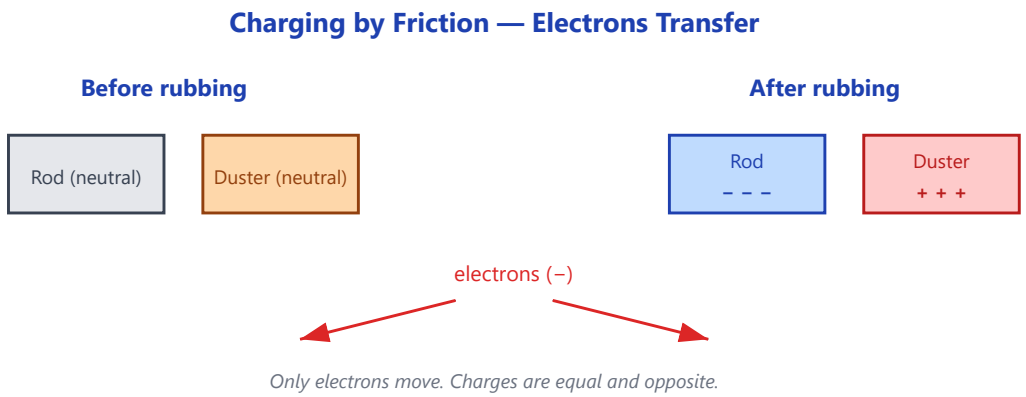
P3 — Electricity and Magnetism

OCR GCSE Physics A (J249) — Gateway Science · Paper 1

P3.1 Static Electricity

> **Static electricity** is a build-up of electrical charge on an insulator. It is produced by rubbing two insulating materials together — electrons transfer from one to the other, leaving one positively charged and the other negatively charged.

When a polythene rod is rubbed with a duster, electrons are transferred from the duster onto the rod. The rod becomes **negatively charged**; the duster becomes **positively charged**. Only electrons move; the positive protons in the nucleus are fixed.



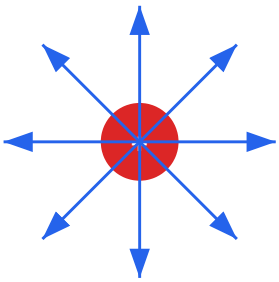
The Law of Charges

- **Like charges repel** (two negatives, or two positives).
- **Unlike charges attract** (a positive and a negative).
- A charged object will **attract** a small uncharged object (by inducing opposite charges on its surface).

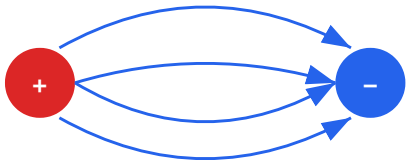
Electric Fields

A charged object creates an **electric field** in the space around it. Another charged object placed in this field will experience a force. Field lines point from **+ to −** and show the direction a positive test charge would move.

Electric Field Lines



Positive: field lines point outwards



Field lines go from + to −

Field strength is shown by how close the lines are together. Close to a charge the lines are close → strong field. Further away the lines spread out → weak field. The field is strongest at the **surface** of a charged object.

Sparks and Earthing

If enough charge builds up on an object, the electric field becomes strong enough to ionise the air around it. Electrons jump across a short gap, producing a visible **spark**.

To stop charge building up we use **earthing** — a wire connects the object to the ground, so any charge flows safely away before it builds up enough to spark. This is why fuel tankers are earthed before unloading petrol (a spark could ignite the vapour).

P3.2 Electric Current & Circuits

An **electric current** is the flow of charge. In a metal wire the charge carriers are free electrons. Current is measured in **amperes (A)** using an **ammeter** placed in series. 1 A = 1 coulomb of charge per second.

$$\mathbf{Q = I \times t} \quad (C) = (A) \times (s)$$

Q = charge, I = current, t = time.

Potential Difference (Voltage)

Potential difference (V), or voltage, is the energy transferred per unit charge. 1 volt = 1 joule per coulomb. Measured with a **voltmeter** placed in parallel across a component.

$$\mathbf{E = Q \times V} \quad (J) = (C) \times (V)$$

Energy transferred by a charge = charge × potential difference.

Resistance & Ohm's Law

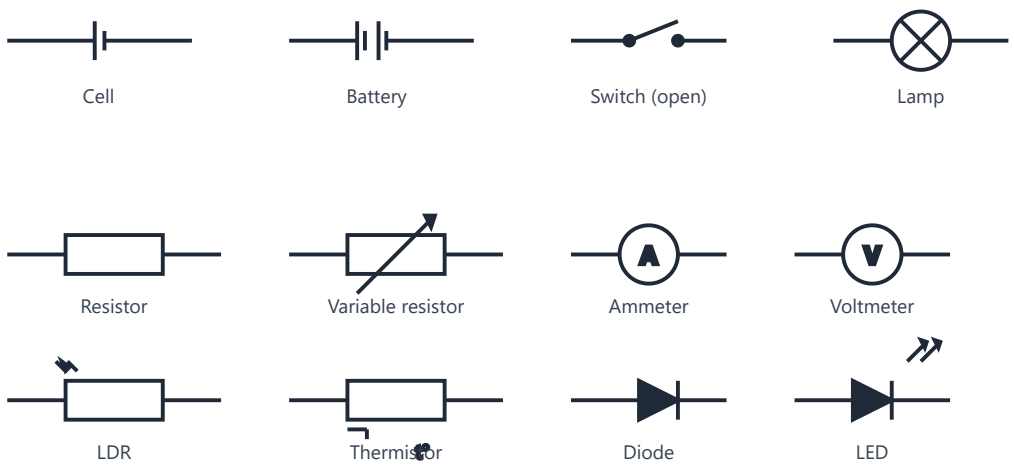
Resistance (R) is how much a component opposes the flow of charge. Higher R → smaller current for the same pd. Unit: ohm (Ω).

$\mathbf{V = I \times R}$ (V) = (A) × (Ω) — *Ohm's law*, for fixed-R components at constant temperature.

Worked Example: A 12 V battery drives a current of 0.4 A through a lamp. Resistance?
R = V / I = 12 / 0.4 = **30 Ω**.

Circuit Symbols

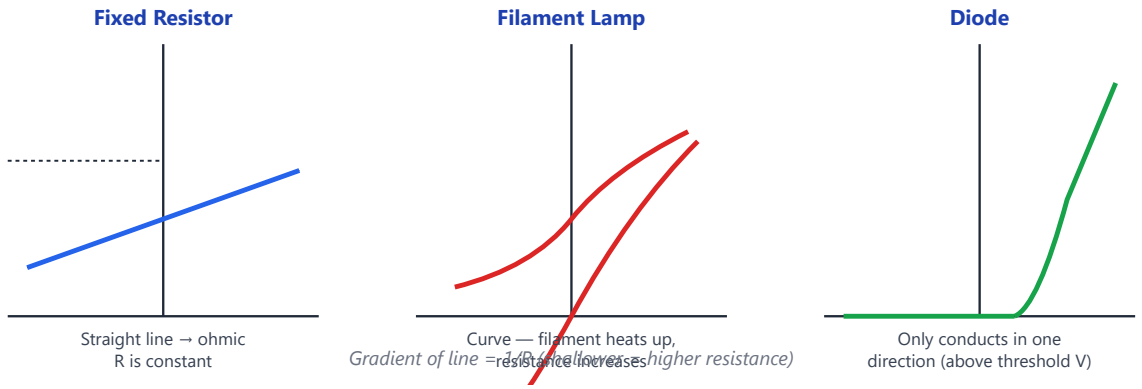
Standard Circuit Symbols



I-V Characteristics

The relationship between current and potential difference tells us how a component behaves. Each component has a distinctive I-V graph.

I–V Characteristics of Three Components

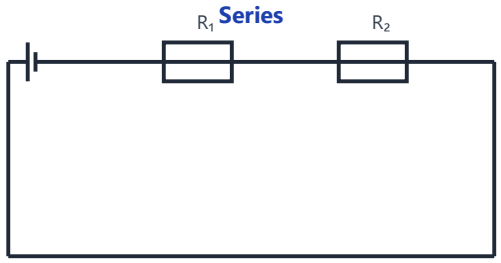


Thermistors and LDRs (Sensors)

Sensor	Resistance behaviour	Use
LDR (Light-Dependent Resistor)	Resistance **decreases** as light intensity increases.	Automatic street lights, camera light meters
Thermistor	Resistance **decreases** as temperature increases (NTC type).	Thermostats, car engine sensors, fire alarms

P3.3 Series and Parallel Circuits

Series and Parallel Circuits



Same current everywhere
$V_1 + V_2 = V$

$_{total}$ $R_{total} = R_1 + R_2$ Parallel R_1 R_2 Same pd across each branch $I_{total} = I_1 + I_2$ R_{total} < smallest branch R

Series Circuits

- Single loop — components one after another.
- **Current is the same everywhere** (no branches to split into).
- **Potential difference adds up**: total pd across components = battery pd.
- **Total resistance = sum of resistances**.
- If one bulb breaks, all go off.

Parallel Circuits

- Branches — components on separate loops.
- **Pd is the same across each branch** (each is directly across the battery).
- **Current splits between branches**: total current = sum of branch currents.
- Total resistance is **less than** the smallest individual resistance — more paths = easier flow.
- If one branch breaks, the others keep working. This is why household lighting is wired in parallel.

Worked Example: Two 6 Ω resistors in series: $R_{total} = 6 + 6 =$ **12 Ω**.

© RAG Learning · Not for redistribution

Two 6 Ω resistors in parallel: R_{total} = (6 × 6) / (6 + 6) = 36/12 = **3 Ω** (half of one resistor).

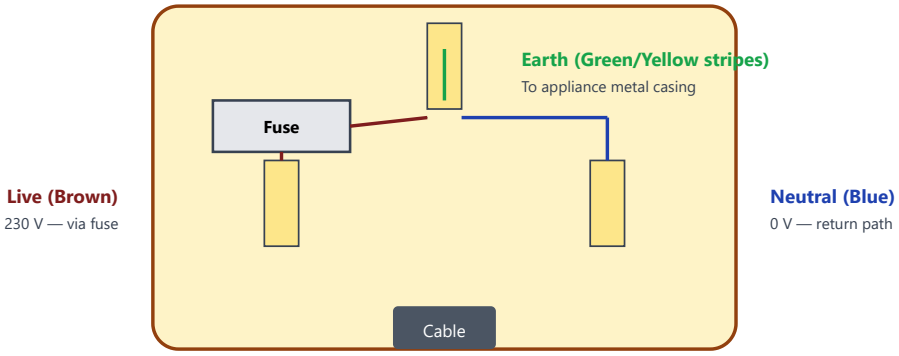
P3.4 Mains Electricity & Power

dc vs ac

Direct current (dc) flows in one direction only — from cells and batteries.
Alternating current (ac) reverses direction many times per second — UK mains is ac, 230 V at 50 Hz.

The Three-pin Plug

UK Three-pin Plug (Opened)



Wire	Colour	Function
Live	Brown	Carries alternating pd of 230 V from mains. Dangerous.
Neutral	Blue	Near 0 V — completes the circuit back to the local substation.
Earth	Green & yellow stripes	At 0 V. Connected to metal casing — safety wire to prevent electric shock if a fault connects the live to the casing.

Fuses and Circuit Breakers

© RAG Learning · Not for redistribution

If a fault causes the live wire to touch the earthed metal casing, a **large current** flows through the earth wire. This current melts the **fuse** (or trips a **circuit breaker**) which breaks the live connection, making the appliance safe.

- Fuses are rated (3 A, 5 A, 13 A). Choose one slightly above the appliance's normal operating current.
- Circuit breakers trip faster and are resettable.

Electrical Power

$$\mathbf{P = V \times I} \quad (\mathrm{W}) = (\mathrm{V}) \times (\mathrm{A})$$

$$\mathbf{P = I^2 \times R} \quad \text{(when R is known)}$$

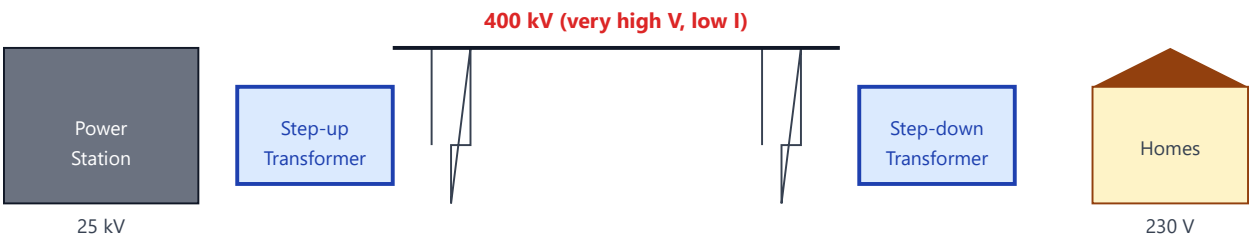
$$\mathbf{E = P \times t} \quad \text{(energy transferred, J)}$$

Worked Example: A kettle is rated 2 300 W at 230 V. Current drawn?
I = P / V = 2300 / 230 = **10 A**. Choose a 13 A fuse (the next standard rating above 10).

The National Grid & Transformers

Power is sent across the country through the **National Grid**. To minimise energy losses in the cables, power is transmitted at very high voltage and low current. Transformers step voltage up for transmission and step it down for homes.

National Grid — Stepping Up, Stepping Down



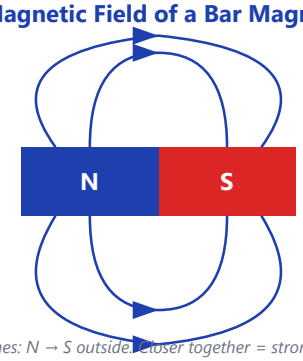
Why high voltage? Power loss in cables = I^2R. Sending at high V means low I, so I^2 loss is tiny. A small increase in V makes a big reduction in energy wasted as heat.

P3.5 Magnetism and the Motor Effect

Permanent Magnets and Magnetic Fields

Every magnet has a **north (N)** and **south (S)** pole. **Like poles repel, unlike poles attract** — the same rule as electric charges. Field lines run **from N to S outside the magnet**.

Magnetic Field of a Bar Magnet



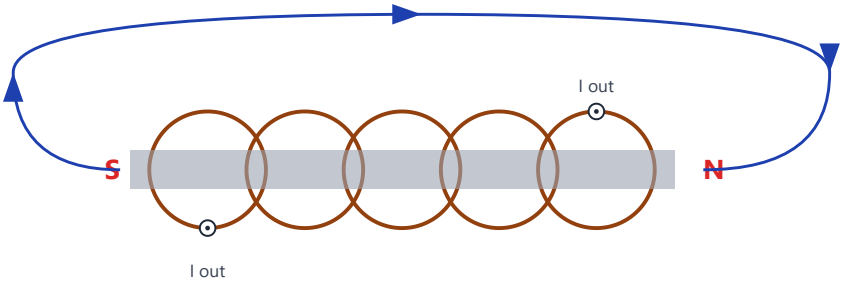
Field lines: N → S outside. Closer together = stronger field.

Induced magnets: iron, cobalt and nickel become magnetic when placed in a magnetic field, but lose most of their magnetism when the field is removed. **Hard magnetic materials** (like steel) keep their magnetism — these are used for permanent magnets.

Electromagnets — Magnetic Fields from Current

A current in a wire creates a magnetic field around it. For a straight wire the field lines form circles; use the **right-hand grip rule** (thumb along current, fingers curl around in field direction).

Solenoid — A Coil Acts Like a Bar Magnet



Adding an iron core increases the field strength many times — that's an electromagnet.

Uses of electromagnets: scrapyard cranes (switch off to drop the load), electric bells, relays, MRI scanners, maglev trains, earphones/loudspeakers.

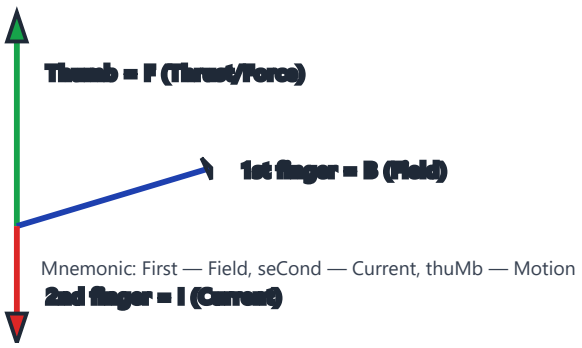
The Motor Effect

When a **current-carrying wire is placed in a magnetic field**, it experiences a force. This is called the **motor effect**. Use **Fleming's left-hand rule** to work out the direction.

$$\mathbf{F = B \times I \times l} \ \ (\mathrm{N}) = (\mathrm{T}) \times (\mathrm{A}) \times (\mathrm{m})$$

B = magnetic flux density (T), I = current (A), l = length of wire in field (m).

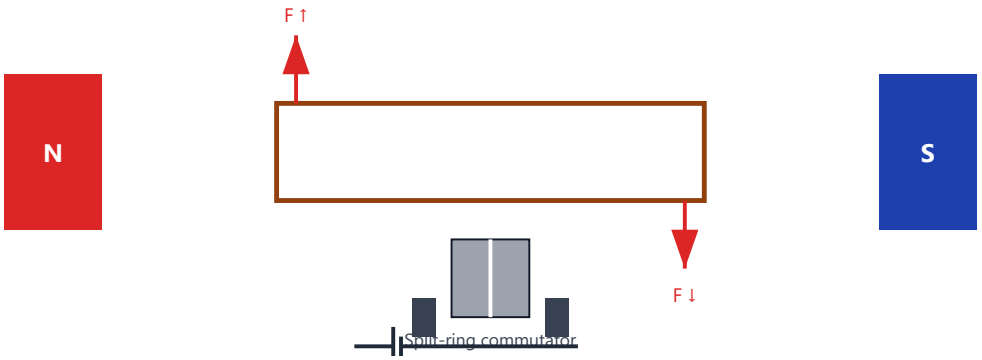
Fleming's Left-Hand Rule



The dc Motor

A coil of wire carrying a current in a magnetic field experiences a force on each side (opposite directions), which makes it spin. A **split-ring commutator** reverses the current in the coil every half turn, so the forces keep turning it the same way.

Simple DC Motor



To **increase the force** (so the motor spins faster): increase current, increase magnetic field strength, or add more turns to the coil.

Quick Summary — Key Equations for P3

Quantity	Equation	Units
Charge flow	$Q = I\,t$	C
Energy transfer (elec.)	$E = Q\,V$	J
Ohm's law	$V = I\,R$	V / A / Ω
Power (general)	$P = V\,I$	W
Power (with R)	$P = I^2\,R$	W
Energy (mains)	$E = P\,t$	J
Motor-effect force	$F = B\,I\,l$	N

© RAG Learning · Not for redistribution